

Option B: Connect the AI Loan Recovery Agent with VICIdial

Production architecture using VICIdial, Asterisk ARI ExternalMedia, RTP media bridge, Sarvam STT/TTS, and Gemini

Prepared for the Loan Recovery Voice Agent project

1. Executive Summary

Option B means VICIdial remains the dialer and Asterisk handles the live call, while the AI agent becomes a server-side telephony bot. The customer audio must be bridged from Asterisk to your AI backend. The browser mic/speaker page can still be used for testing, but production VICIdial calling requires an Asterisk media bridge.

Component	Role
VICIdial	Campaigns, lead management, dialing, dispositions, and agent/admin UI.
Asterisk	Telephony layer underneath VICIdial; answers calls and routes audio.
ARI ExternalMedia	Creates an external media channel so audio can be sent to a custom service.
RTP Media Bridge	Receives caller audio, performs VAD/chunking, sends bot audio back.
FastAPI Agent Backend	Owns RecoveryAgentSession, customer data, PTP logic, logs, and outcomes.
Sarvam STT/TTS	Speech-to-text and text-to-speech for live conversations.
Gemini	Dynamic customer-facing response generation from policy/workflow guidance.

2. Production Architecture

Target call flow:

```
Customer phone
  <-> VICIdial / Asterisk
  <-> ARI ExternalMedia channel
  <-> RTP media bridge
  <-> FastAPI AI agent session
  <-> Sarvam STT -> Gemini -> Sarvam TTS
  <-> RTP media bridge
  <-> Asterisk
  <-> Customer hears Priya
```

Asterisk ARI ExternalMedia is the preferred production route because it is designed to direct media from an Asterisk channel to an external media service. The official Asterisk documentation describes ExternalMedia as a way to send media to a proxy service, such as a speech-recognition or custom media processor.

3. Step 1 - Keep the Current AI Backend Running

On the AI server, start the FastAPI backend with Gemini and Sarvam configured:

```
cd /path/to/loan_recovery_agent
```

```
export GEMINI_API_KEY="your_gemini_key"
export SARVAM_API_KEY="your_sarvam_key"
export GEMINI_MODEL="gemini-3.5-flash"
export AGENT_USE_DYNAMIC_LLM=1
export AGENT_STRICT_DYNAMIC_LLM=1
export AGENT_DEBUG_LOGS=1
```

```
python -m uvicorn web_voice_server:app --host 0.0.0.0 --port 8000
```

Health check:

```
curl http://AI_SERVER_IP:8000/health
```

Expected response:

```
{"ok": true}
```

4. Step 2 - Add Telephony Session APIs

The browser WebSocket path should remain for testing. For VICIdial, add server-side session endpoints so the Asterisk bridge can create and advance a call session without using browser audio.

4.1 Start Session Endpoint

```
POST /vicidial/session/start
```

Example request:

```
{
  "lead_id": "12345",
  "phone_number": "9991110001",
  "first_name": "Rahul Patel",
  "loan_id": "701",
  "due_amount": 2500,
  "loan_date": "2026-05-11",
  "dealer_name": "Dealer One",
  "PTP_bucket": "0",
  "Max_ptp_extension": 3,
  "asterisk_uniqueid": "1749550000.12",
  "campaign_id": "AI_RECOVERY"
}
```

The endpoint should normalize this into the same customer format already used by the agent and then call `session.start()`.

```
customer_record = normalize_frontend_customer(payload)
session = RecoveryAgentSession(customer_record, all_customers=None)
result = session.start()
return {"ok": True, "call_id": call_id, "bot_text": result["bot_text"]}
```

4.2 Turn Endpoint

```
POST /vicidial/session/{call_id}/turn
```

Example request:

```
{"transcript": "हाँ मैं राहुल बोल रहा हूँ"}
```

Example response:

```
{
  "bot_text": "राहुल जी, आपकी 2500 रुपये की EMI pending है...",
  "should_end_call": false,
}
```

```

    "outcome": null,
    "state": {}
}

```

5. Step 3 - Configure ARI on Asterisk/VICIdial

On the VICIdial/Asterisk server, enable ARI HTTP and create an ARI user.

5.1 /etc/asterisk/http.conf

```

[general]
enabled=yes
bindaddr=0.0.0.0
bindport=8088

```

5.2 /etc/asterisk/ari.conf

```

[general]
enabled=yes
pretty=yes

[aiagent]
type=user
read_only=no
password=StrongPasswordHere

```

5.3 Reload and Test

```

asterisk -rx "core reload"
asterisk -rx "http show status"

```

```

curl -u aiagent:StrongPasswordHere http://ASTERISK_SERVER_IP:8088/ari/asterisk/info

```

6. Step 4 - Route Calls into an ARI App

Add a custom dialplan context. In many deployments this is placed in /etc/asterisk/extensions_custom.conf.

```

[ai-recovery]
exten => 860001,1,NoOp(AI Recovery Bot)
same => n,Answer()
same => n,Stasis(ai_recovery_app,${CALLERID(num)},{UNIQUEID})
same => n,Hangup()

```

Reload the dialplan:

```

asterisk -rx "dialplan reload"

```

7. Step 5 - Build the ARI Call-Control Bridge

Create a new service separate from web_voice_server.py:

```

vicidial_bridge/ari_app.py

```

Responsibilities:

- Connect to the Asterisk ARI WebSocket.
- Listen for StasisStart events for ai_recovery_app.
- Read caller number, Asterisk channel ID, and uniqueid.
- Start an AI session using /vicidial/session/start.

- Create a mixing bridge in Asterisk.
- Add the customer channel to the bridge.
- Create an ExternalMedia channel pointed at your RTP media server.
- Add the ExternalMedia channel to the same bridge.
- On final AI outcome, hang up, transfer, or update VICIdial status.

```
StasisStart(customer channel)
  -> create AI call session
  -> create Asterisk bridge
  -> add customer channel
  -> create ExternalMedia channel
  -> add ExternalMedia channel
  -> RTP media server handles audio
```

8. Step 6 - Build the RTP Media Server

Create the real-time media component:

```
vicidial_bridge/rtp_media_server.py
```

Responsibilities:

- Receive RTP audio packets from Asterisk.
- Decode to PCM16 mono audio, ideally 16 kHz for easiest STT integration.
- Run VAD/utterance detection so STT is called only after a customer utterance ends.
- Send completed speech audio to Sarvam STT.
- Send transcript to /vicidial/session/{call_id}/turn.
- Send returned bot_text to Sarvam TTS.
- Convert bot audio into the codec/rate expected by Asterisk.
- Send RTP audio back into the ExternalMedia channel.

Recommended Media Setting	Value
Codec for first prototype	slin16 / PCM16 where available
Sample rate	16000 Hz mono
VAD trailing silence	400-700 ms
First goal	Receive-only RTP packet/RMS logging before adding STT

9. Step 7 - Map VICIdial Lead Data to Agent Customer Data

Choose one lead-data lookup method.

9.1 Method A - Lookup by Phone in Your Loan Backend

```
GET http://AI_SERVER:8000/customer/by-phone/9991110001
```

9.2 Method B - Lookup by VICIdial Lead ID

Use VICIdial database/API fields and map them into the agent customer object:

VICIdial Field	AI Agent Field
lead_id	lead_id
phone_number	phone_number
first_name	first_name
vendor_lead_code	loan_id
custom due amount field	due_amount
custom PTP bucket field	PTP_bucket
custom max PTP field	Max_ptp_extension

comments or custom dealer field	dealer_name
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10. Step 8 - Update VICIdial After the AI Call

When the AI session returns a final outcome, update VICIdial lead status and your customer record.

AI Outcome	Suggested VICIdial Status
PTP_CREATED	PTP
COMMITTED	PROMISE
WRONG_PERSON	WN
ESCALATED	ESCALATE
CUSTOMER_ENDED	CALLBK or NI
CALL_ABANDONED	DROP or NA

Example Non-Agent API style call:

```
curl "http://VICIDIAL_SERVER/vicidial/non_agent_api.php?
source=ai_agent&user=APIUSER&pass=APIPASS&function=update_lead&lead_id=12345&status=PTP&comments=AI%20PTP%20created"
```

11. Step 9 - Test in This Exact Order

1. Test the AI backend alone using /health and the browser /call page.
2. Add and test /vicidial/session/start with curl.
3. Add and test /vicidial/session/{call_id}/turn with typed transcripts.
4. Call extension 860001 from a softphone and verify ARI receives StasisStart.
5. Create ExternalMedia and verify the RTP server receives packets.
6. Log RMS levels and confirm levels change when the caller speaks.
7. Add Sarvam STT and verify customer transcripts.
8. Add agent turn processing and verify bot_text returns.
9. Add Sarvam TTS playback back into the Asterisk call.
10. Update VICIdial final status and verify lead disposition.
11. Run one test lead from VICIdial.
12. Only after stable tests, run a small batch of 10-20 leads.

12. Latency Targets and Optimization

Phase	Target
VAD after customer stops	400-700 ms
STT	700-1500 ms
LLM	500-1500 ms
TTS	800-2000 ms
Total turn latency	2.5-5.0 seconds

If latency is high, reduce LLM retries, shorten bot responses, reduce VAD silence, avoid repeated policy explanations, and ensure only one LLM call is made per customer turn.

13. Recommended Folder Layout

```
loan_recovery_agent/
├── agent_core.py
├── web_voice_server.py
```

```

├── voice_io.py
├── config.py
├── vicidial_bridge/
│   ├── ari_app.py
│   ├── rtp_media_server.py
│   ├── vicidial_api.py
│   ├── audio_utils.py
│   └── README.md
├── logs/
├── dynamic_call_records/
├── frontend/
│   └── call.html

```

14. Minimum Environment Variables

```

GEMINI_API_KEY=your_key
SARVAM_API_KEY=your_key
GEMINI_MODEL=gemini-3.5-flash
AGENT_USE_DYNAMIC_LLM=1
AGENT_STRICT_DYNAMIC_LLM=1
AGENT_DEBUG_LOGS=1

VICIDIAL_BASE_URL=http://VICIDIAL_SERVER
VICIDIAL_API_USER=api_user
VICIDIAL_API_PASS=api_password

ASTERISK_ARI_URL=http://ASTERISK_SERVER:8088/ari
ASTERISK_ARI_USER=aiagent
ASTERISK_ARI_PASS=StrongPasswordHere
AI_SERVER_URL=http://AI_SERVER:8000

```

15. Production Checklist

- Use HTTPS/VPN or private networking between VICIdial, Asterisk, and the AI server.
- Do not expose ARI publicly without firewall restrictions.
- Store Gemini, Sarvam, VICIdial, and ARI credentials in environment variables, not Git.
- Start with one call at a time until the RTP bridge is stable.
- Log every call_id, uniqueid, lead_id, transcript, latency, outcome, and PTP result.
- Add human escalation routing before running live production traffic.
- Confirm compliance and consent requirements for AI voice calls and recording in your operating region.

16. Recommended Implementation Phases

Phase	Goal	Success Criteria
1	Stabilize browser agent	Gemini, Sarvam, PTP flow, and logs work through /call.
2	Add telephony session APIs	curl can start a call and send transcript turns.
3	Build ARI call-control app	Softphone call enters Stasis and logs channel/uniqueid.
4	Build RTP receive path	ExternalMedia sends packets and RMS changes with speech.
5	Connect STT and agent turn	Customer speech produces bot_text.
6	Connect TTS playback	Customer hears Priya through the phone call.
7	Update VICIdial status	PTP/commit/escalation outcomes update the lead.
8	Run AI campaign pilot	Small controlled batch with monitoring

		and escalation.
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17. References

Official Asterisk ARI ExternalMedia documentation:

<https://docs.asterisk.org/Development/Reference-Information/Asterisk-Framework-and-API-Examples/External-Media-and-ARI/>

Official VICIdial Non-Agent API documentation:

https://vicidial.org/docs/NON-AGENT_API.txt